

Marcus Gafka

msgafka@gmail.com | [linkedin.com/in/marcus-gafka](https://www.linkedin.com/in/marcus-gafka)

OBJECTIVE

Passionate Robotics and Mechanical Engineering senior graduating May 2027, seeking a full-time position developing innovative robotic systems. Eager to contribute and expand hands-on experience in mechanical design, controls, system integration, testing, and solving complex engineering problems.

EDUCATION

- 8/2023 – present** **Worcester Polytechnic Institute (Worcester, MA)**
- **Full Scholarship:** Single recipient of WPI full-tuition, merit-based FRC scholarship
 - **Major:** B.S. Robotics Engineering & B.S. Mechanical Engineering
 - **GPA: 3.94/4.0, Graduation: May 2027**
 - **Coursework Including:** Robotics, Computer Vision, Motion Planning, Control Systems, Software Design, Embedded Computing, Integrated Mechanical Design
 - **Study Abroad:** Completed a quarter-long “Buildings of Venice, Italy” project
- 8/2019 – 5/2023** **Clear Lake High School (Houston, TX)**
- **Top 1.7%** (Rank: 10/605), **Weighted +5.0+ GPA:** 5.6861
 - **SAT:** 1530 (Superscore: Math = 800, Reading = 730)
 - Rigorous schedule including: 11 AP classes, 4 Dual Credit, many Pre-AP/Honors
 - Captained multiple highly competitive robotics teams: FRC118 and 2373M (VEX)

ROBOTICS INTERNSHIP EXPERIENCE

- Summer 2026** **Controls Engineering Intern – DEKA Research & Development (Manchester, NH)**
- Developed an Arduino-based platform with magnetic encoder feedback to validate closed-loop stepper motor control for testing prototypes
 - Developed Python GUI-controlled software and automated Bode plot generation from real-time data to characterize frequency response of nonlinear system
 - Diagnosed sensor limitations including noise, resolution, and system latency; improved control performance using tailored Kalman filters and hysteresis logic
- Summer 2025** **NASA Robotics Academy Intern / University Intern – NASA JSC (Houston, TX)**
- 2024**
- 2023**
- Led construction of high-fidelity lunar rover mock-up, assembled rover chassis, electronic housings, operator interfaces, and differential-swerve modules
 - Maintained Space Exploration Vehicle and led design of a new door latch mechanism
 - Led design and manufacturing process of dynamic trophies for Space City VEX Event

ACTIVITIES

- 8/2023 – present** **FRC190 College Mentor – WPI (approx. 400 hr/year + travel for competitions)**
- Mentoring high-school students in FIRST Robotics Competition (FRC), guiding them through complex engineering projects and developing technical and leadership skills.
 - Current drive coach, tasked with strategic planning and inter-team coordination at competitions and overseeing team execution on the playing field during matches
- 9/2025 – present** **Rho Beta Epsilon Robotics Honor Society – WPI – Elected New Member Officer**
- Leading recruitment and induction for prospective members
 - Developing technical questions, rubrics, and interview processes
 - Coordinating applicant interviews and member evaluation

SKILLS

CAD	Software: PTC Creo, Solidworks (with PDM), Onshape, Autodesk Inventor Processes: Master sketching, Master modeling, Parametric design, Complex assemblies
CAM/CNC	Autodesk Fusion for: HAAS VF4-SS CNC Mill, Shop Sabre 4x4 ft CNC table router
3D printing	Various slicers for 3d printing on Prusa, Bambu, Markforged, and Formlabs machines
Machining	Lathe, Mill, Drill Press, Vertical Bandsaw, Horizontal Bandsaw, Powered hand tools
Software	Linux, ROS/ROS2, Git, MatLab, Arduino, Python, C++, C, Arduino, Raspberry Pi, Java